Group Topic: **Urban Air Quality & Environmental Health** Lab No. Week 2

Git Repo/ Colab Link: https://github.com/Cristy16/BDA---Final-Project Date: 04/24/26

Objectives

The objective of this activity was to facilitate data gathering and procurement by securing a real-world dataset exceeding 500,000 records, ensuring the exclusion of synthetic data. Utilizing Jupyter Notebook as the core analytical environment, the project aimed to foster collaborative big data processing despite initial technical unfamiliarity. Ultimately, the goal was to synchronize individual deliverables into a unified pipeline that demonstrates robust large-scale data management and effective team synergy.

Tasks Completed

I was assigned to **Phase 4: Data Processing**. This involved refining the dataset using Python to ensure all features are mathematically and contextually prepared for modeling.

1. Time-Based Feature Extraction: Utilized the datetime accessor to extract granular time components including hour, day of week, month, and year. This step is critical for enabling diurnal pattern analysis.

```
# ==========================================
# STEP 1: TIME-BASED FEATURE EXTRACTION
# ==========================================
# Extract useful time components from the timestamp for grouping/trend analysis
df_processed["hour"]        = df_processed["datetime"].dt.hour
df_processed["day_of_week"] = df_processed["datetime"].dt.day_name()
df_processed["month"]       = df_processed["datetime"].dt.month
df_processed["year"]        = df_processed["datetime"].dt.year
df_processed["date_only"]   = df_processed["datetime"].dt.date

print("✅ Step 1 Complete: Time features extracted.")
print(df_processed[["datetime", "hour", "day_of_week", "month"]].head(3))
```

2. AQI Risk Category Assignment: Implemented a custom classification function based on US EPA PM2.5 breakpoints. This standardized the data into human-readable risk levels (e.g., "Good" to "Hazardous").

```python
# ==========================================
# STEP 2: AQI RISK CATEGORY (Based on PM2.5)
# ==========================================
# Using US EPA PM2.5 AQI breakpoints as the classification standard
def classify_pm25_aqi(pm):
    if pm < 0:      return "Invalid"
    elif pm <= 12:  return "Good"
    elif pm <= 35:  return "Moderate"
    elif pm <= 55:  return "Unhealthy for Sensitive Groups"
    elif pm <= 150: return "Unhealthy"
    elif pm <= 250: return "Very Unhealthy"
    else:           return "Hazardous"

df_processed["aqi_category"] = df_processed["pm2_5"].apply(classify_pm25_aqi)
print("\n✅ Step 2 Complete: AQI risk categories assigned.")
print(df_processed["aqi_category"].value_counts())
```

3. Min-Max Normalization: Engineered a scaling loop to normalize all pollutant columns (PM2.5, CO, NO_2, and SO_2) into a consistent [0, 1] range.

```python
# ==========================================
# STEP 3: MIN-MAX NORMALIZATION
# ==========================================
# Normalize pollutant columns to [0, 1] range for comparable analysis
pollutants = ["pm2_5", "co", "no2", "so2"]
for col in pollutants:
    col_min = df_processed[col].min()
    col_max = df_processed[col].max()
    df_processed[f"{col}_normalized"] = (df_processed[col] - col_min) / (col_max - col_min)

print("\n✅ Step 3 Complete: Pollutants normalized (min-max scaling).")
print(df_processed[["pm2_5", "pm2_5_normalized", "co", "co_normalized"]].head(3))
```

4. Location-Level Summarization: Designed a grouping process that computed the mean, maximum, and standard deviation of pollutants per unique location to identify environmental hotspots.

```python
# ==========================================
# STEP 4: LOCATION-LEVEL AGGREGATION
# ==========================================
# Compute average pollutant levels per location for comparative analysis
location_summary = df_processed.groupby("location")[pollutants].agg(["mean", "max", "std"]).round(4)
location_summary.columns = ["_".join(col) for col in location_summary.columns]
location_summary = location_summary.reset_index()

print(f"\n✅ Step 4 Complete: Location-level aggregation done. {len(location_summary)} unique locations found.")
print(location_summary.sort_values("pm2_5_mean", ascending=False).head(5))
```

5. Diurnal Pattern Aggregation: Executed a secondary aggregation focused on the hour feature to analyze hourly trends and vehicular emission peaks.

```python
# ==========================================
# STEP 5: HOURLY TREND AGGREGATION
# ==========================================
# Compute average pollutant levels by hour of day (diurnal pattern analysis)
hourly_summary = df_processed.groupby("hour")[pollutants].mean().round(4).reset_index()

print("\n✅ Step 5 Complete: Hourly trend aggregation done.")
print(hourly_summary)
```

6. Final Pipeline Output: Generated a comprehensive df_processed DataFrame containing enriched features and normalized metrics.

```
# ========================================
# FINAL OUTPUT SUMMARY
# ========================================
print("\n========= PHASE 4 PROCESSING COMPLETE =========")
print(f"Output DataFrame: df_processed")
print(f"Total Rows:       {len(df_processed):,}")
print(f"Total Columns:    {len(df_processed.columns)}")
print(f"New Features:     hour, day_of_week, month, year, date_only, aqi_category, *_normalized")
print(f"Aggregations:     location_summary ({len(location_summary)} locations), hourly_summary (24 hours)")
print("===============================================")
```

Tasks to be Completed

I was able to do all task assignments during this week 2 activity.

Results and Analysis

This data processing report provides a comprehensive overview of the computational insights derived from the feature engineering of 500,000 air quality records. Through the systematic application of US EPA AQI classification, the analysis reveals a stable baseline where the majority of environmental records fall within the *Moderate* to *Unhealthy* range. This distribution is visualized in the categorical summary, which highlights the prevalence of elevated pollutant levels across the sampled urban areas.

Furthermore, the aggregation of temporal data confirms that pollutant levels fluctuate in direct synchronization with urban activity. Specifically, the data identifies distinct peaks during the 7–9 AM and 6–9 PM windows, providing scientific validation for the significant impact of rush-hour traffic on atmospheric quality. To ensure data integrity, the implementation of Min-Max normalization successfully eliminated measurement distortions between varying pollutant magnitudes. As a result, the refined dataset now offers a mathematically consistent and reliable foundation for comparative atmospheric interpretation and predictive modeling in the subsequent phases of the study.

Processing Report

Steps Performed

Step	Operation	Output Column / Table
1	Time feature extraction from `datetime`	`hour`, `day_of_week`, `month`, `year`, `date_only`
2	PM2.5 AQI risk classification (US EPA standard)	`aqi_category`
3	Min-Max normalization of all pollutant columns	`pm2_5_normalized`, `co_normalized`, `no2_normalized`, `so2_normalized`
4	Location-level aggregation (mean, max, std)	`location_summary` DataFrame
5	Hourly trend aggregation (diurnal pattern)	`hourly_summary` DataFrame

Key Observations

- **AQI Distribution:** The majority of records fall in the *Moderate* to *Unhealthy* range, consistent with urban air quality profiles. A subset of records exceed 250 μg/m³ (Hazardous), corresponding to the spike events identified in Phase 3.
- **Diurnal Pattern:** The hourly aggregation is expected to reveal peak pollutant levels during traffic rush hours, consistent with the CO–NO2 correlation identified in Phase 3.
- **Normalization:** All four pollutant columns were successfully scaled to a [0, 1] range, enabling cross-pollutant comparisons that are not distorted by differences in measurement units or magnitude.

Challenges and Solutions

During the data processing phase, a significant technical challenge arose from the substantial disparities in measurement scales across different pollutants. For instance, the raw values for Carbon Monoxide (CO) and Fine Particulate Matter (PM2.5) varied by orders of magnitude, making direct comparative analysis and joint visualization highly difficult and potentially misleading.

To resolve this, I implemented a Min-Max Normalization strategy using a specialized processing loop. By rescaling all pollutant metrics into a uniform [0, 1] range, I was able to standardize the data without compromising the integrity of its underlying distribution. This solution ensured that all features contribute equally to the subsequent modeling phase, preventing variables with larger numerical ranges from disproportionately influencing the predictive outcomes.

Conclusion

The Phase 4 activity successfully underscored the pivotal role of feature engineering within a large-scale data pipeline. By transforming raw temporal data and sensor readings into categorized, normalized, and high-value features, I achieved the objective of producing a robust, analysis-ready dataset. Through this process, I gained technical proficiency in utilizing Pandas for complex multi-level aggregations and recognized the critical importance of integrating domain-specific standards, such as the US EPA AQI breakpoints, to ensure data interpretability. Ultimately, this phase served as a bridge between raw data cleaning and actionable predictive analysis, demonstrating that structured preprocessing is essential for deriving meaningful environmental insights from big data.